Bassaganyas *et al. European Radiology*
https://doi.org/10.1007/s00330-025-11772-1

EUROPEAN SOCIETY OF RADIOLOGY
European Radiology

ULTRASOUND

Role of elastography and dynamic contrast-enhanced ultrasound in the evaluation of pancreas transplantation rejection

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Abstract

Objectives Rejection is the leading cause of graft failure, and its diagnosis remains a challenge. Elastography and dynamic contrast-enhanced ultrasound (DCE-US) are novel non-invasive techniques for quantifying tissue elasticity and perfusion. Their role in pancreas graft rejection has not yet been defined.

Materials and methods From October 16 to January 20, all pancreas transplantations performed at our institution were prospectively studied with elastography and DCE-US at 1 week, 3 weeks, and 12 months post-transplantation. Surveillance biopsies were performed at 3 weeks and 12 months. Elastography and DCE-US were also conducted in all requested biopsies during this period (regardless of the date of transplantation). Patients were categorized according to the biopsy result: normal/rejection. Grafts with other complications were excluded. Cut-off values were established.

Results One hundred twenty-one elastography and 127 DCE-US in 56 patients were included. All parameters showed a high dispersion during the first 90 days post-transplantation. After this period, the rejection group presented higher stiffness (0.97 vs 1.46 m/s, $p < 0.001$) and lower perfusion. The optimal cut-off value for elastography was 1.27 m/s (AUC 0.80), and for DCE-US were: peak enhancement 601 a.u. (AUC 0.67), wash-in AUC 2748 a.u. (AUC 0.70), wash-in rate 118 a.u. (AUC 0.65), wash-in perfusion index 369 a.u. (AUC 0.67), wash-out AUC 5181 a.u. (AUC 0.69) and total AUC 6388 a.u. (AUC 0.68). A combined predictive score showed that alteration of elastography and DCE-US was associated with a 23.2-fold probability of rejection.

Conclusion After the first 90 days post-transplantation, pancreas graft rejection is associated with higher stiffness and lower graft perfusion.

Key Points

Question *Pancreas graft rejection remains a clinical challenge, as there are currently no reliable non-invasive tests for its diagnosis.*

Findings *After the first 90 days post-transplantation, elastography and DCE-US show higher stiffness and lower pancreas graft perfusion in the presence of rejection.*

Clinical relevance *These non-invasive tools, which can be easily integrated into daily routine practice, may be useful in identifying grafts at higher risk of rejection, allowing closer follow-up or early biopsy to establish early rejection treatment, improving graft and patient survival.*

Keywords Pancreas transplantation, Graft rejection, Elasticity imaging techniques, Ultrasonography

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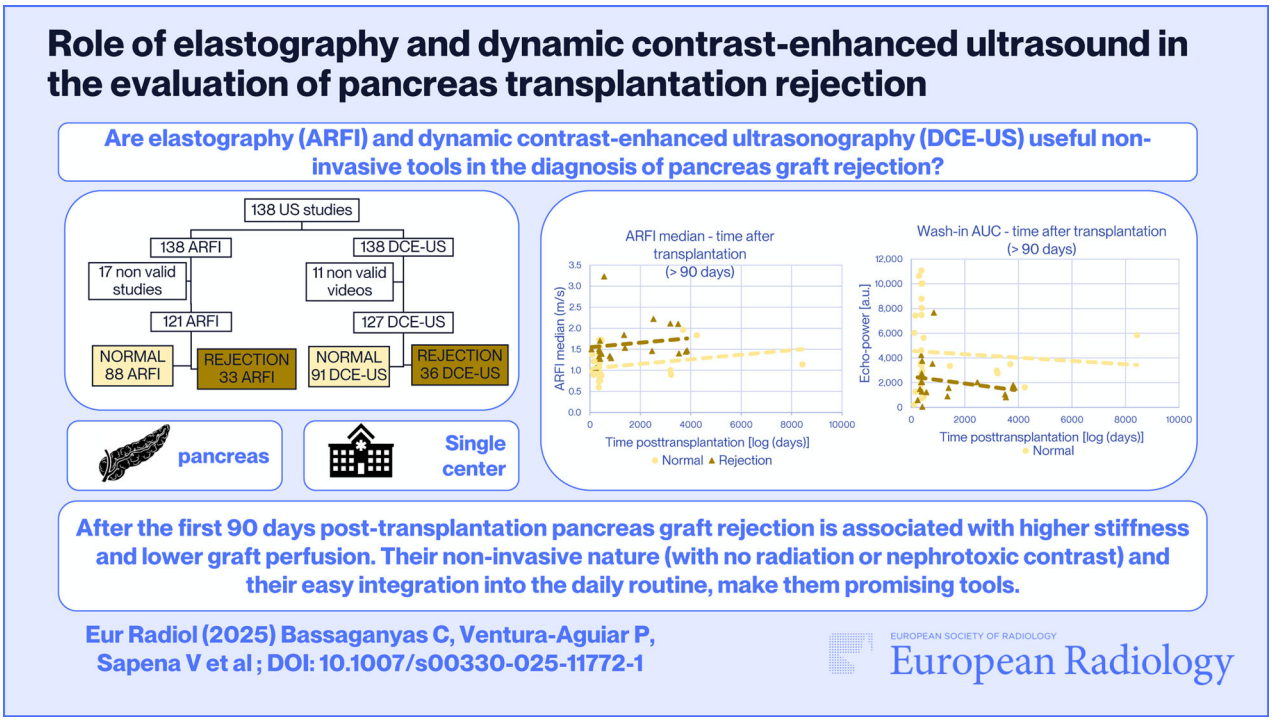
Graphical Abstract



Introduction

Graft rejection is the leading cause of pancreas graft failure beyond the first 90 days after transplantation (20–25% of the grafts) [1], impacting patient survival and quality of life. Accurate and early diagnosis is critical, as early treatment has a significant impact on long-term graft outcomes [1]. However, clinical manifestations and currently available biomarkers may not be altered in subclinical rejection, are non-specific [2], and show an irregular correlation with pancreas graft rejection [3, 4]. Imaging studies are also nonspecific and are mostly used to rule out other causes of graft dysfunction (i.e., graft pancreatitis, vascular complications…) [5–7]. Ultrasonography (US), in expert hands, is usually the first-line imaging technique used to evaluate graft dysfunction, mainly to assess vascular and other postoperative complications [8]. A high resistance index (> 0.7) in the arterial Doppler study has been proposed to predict rejection [9], although its usefulness has not been clearly established [8]. Computed tomography and magnetic resonance imaging can also be used to assess surgical complications, but are not suitable for assessing rejection [5]. Pancreas graft biopsy is therefore the gold standard for diagnosing graft rejection [10]. However, biopsy is not always feasible [11], the graft is not always accessible [12], and its invasive nature entails some risk of complications [13].

Elastography is a non-invasive technique that quantifies the stiffness of the evaluated tissue. It can be performed with US and magnetic resonance elastography (MRE). MRE has been reported to offer higher diagnostic performance and fewer technical limitations (as it is less operator-dependent) [14], but its applicability in the pancreas graft has not yet been investigated. US elastography is more available, less costly, and has fewer acquisition times, resulting in less complexity than MRE. It has been widely used to assess liver pathology, but its application in other organs is limited. In the case of the pancreas, the main limitation is its deep intraabdominal position, with only a few publications on the native pancreas [15]. Some studies in the field of transplantation have reported an increase in stiffness in the presence of rejection in liver [16] and kidney [17] grafts. Unfortunately, there are no data on pancreas transplantation.

Contrast-enhanced ultrasonography (CEUS) is a well-known technique that reflects the state of parenchymal vascularization and is routinely used to assess the vascular permeability of transplanted organs [18]. Sonographic contrast is not nephrotoxic and is very useful in patients with renal dysfunction or a recent kidney transplant.

Quantification of CEUS (DCE-US) through time-intensity curves (TIC) is an emerging technique mainly used to assess tumor perfusion [19, 20], although some authors have tried to apply it in transplanted organs. A few studies in kidney grafts [21–24] and, to our knowledge, only one in pancreas transplantation [25], have shown lower perfusion in grafts with rejection.

This study aims to prospectively evaluate the role of these two non-invasive US techniques, elastography and DCE-US, in the diagnosis of pancreas graft rejection.

Materials and methods

Study population

The study was approved by the Institutional Ethics Committee for Clinical Research of our hospital (Reg. HCB/2016/0479). Informed consent was obtained from all patients prior to their inclusion.

This prospective study included all patients who underwent pancreas transplantation at our center between October 2016 and January 2020. All grafts were evaluated by elastography and DCE-US at 1 week, 3 weeks, and 12 months after transplantation. A surveillance biopsy was performed immediately after the 3-week and 12-month studies. In addition, all patients who were asked to undergo a pancreas graft biopsy during the study period and outside of the surveillance program (regardless of their transplantation date) were also included. These grafts were first evaluated with elastography and DCE-US, and after that, the biopsy was performed. These clinically indicated biopsies were prescribed as follows: (1) graft dysfunction (increase in serum amylase and/or lipase tripling normal value, hyperglycemia, or presence of de novo donor-specific antibodies) and (2) follow-up of a previously treated rejection episode, 4 weeks after completion of treatment [26]. Grafts with complications other than rejection at the time of the US study were excluded from the study.

Patient demographics, donor age, cold ischemia time, time after transplantation ("graft age"), and indication for the study (surveillance/dysfunction/rejection follow-up) were recorded.

Ultrasonographic study

US imaging was performed with an Acuson S3000 Helx (Siemens®), using convex (1–4.5 MHz) and linear (4–9 MHz) multifrequency probes. All studies were performed by three senior radiologists with more than 10 years of experience in US and interventional US. All examinations were performed with the patient resting in the supine position for at least 15 min and fasting for more than 6 h.

After a complete US study of the graft, including Doppler assessment of vascular patency, the elastographic study was performed using shear wave elastography with Acoustic Radiation Force Impulse (ARFI). The study was carried out with no abdominal compression at the graft parenchyma area to be biopsied (usually the site with the greatest parenchyma thickness), using the convex transducer and the "Virtual Touch™ tissue quantification" system. A minimum of seven valid determinations were registered [27], obtaining the median value in m/s. As a quality criterion, those studies with a ratio interquartile range (IQR)/median > 0.3 were excluded [28, 29]. Finally, a 2.4 mL bolus of US contrast agent (Sonovue, Bracco®) was injected into a peripheral vein of the arm and flushed with 10 mL of saline solution. A 2-min video was recorded with the convex probe at the same site where the elastography was previously performed, starting at the time of injection. The videos were stored and later exported in DICOM format to a workstation equipped with the VueBox software (Bracco®) [30, 31] for post-processing. A circular 2 cm^2 region of interest was manually placed in the parenchyma, avoiding large vessels, to obtain the TIC, as shown in Fig. 1. Built-in motion compensation was applied when necessary, to compensate for in-plane motion. The quality of fit (QoF, %) of the obtained TICs was recorded as recommended by the current guidelines [19]. Our DCE-US studies were considered reliable if the QoF was ≥ 65%. Eleven TIC parameters were obtained for each DCE-US study: peak enhancement (PE, a.u.), wash-in area under the curve (WiAUC, a.u.), rise time (RT, s), mean transit time (mTTI, s), time to peak (TTP, s), WiR (a.u.), wash-in perfusion index (WiPi = WiAUC/RT, a.u.), wash-out area under the curve (WoAUC, a.u.), wash-in and wash-out area under the curve (WiWoAUC, a.u.), fall time (FT, s) and wash-out rate (WoR, a.u.), as shown in Fig. 1. A detailed explanation of all studied parameters is included in the Supplementary Material.

Percutaneous graft biopsy and pathologic diagnosis

In the previously indicated cases, a percutaneous US-guided biopsy was performed in the studied area immediately after the US examination using an automatic 18-gauge needle according to the previously described technique [12].

Fresh graft biopsies were immediately sent to the pathology department. After formalin fixation and paraffin embedding, hematoxylin and eosin stains were performed for histomorphological analysis, and immunohistochemical staining with antibodies CD3, CD68, insulin, glucagon, C4d, cytomegalovirus, in situ hybridization for Epstein–Barr virus, as well as histochemical staining with Masson's trichrome for the fibrous component were performed. All samples were analyzed by a single senior pathologist (M.C.). Samples were considered adequate for evaluation if they were sufficient to

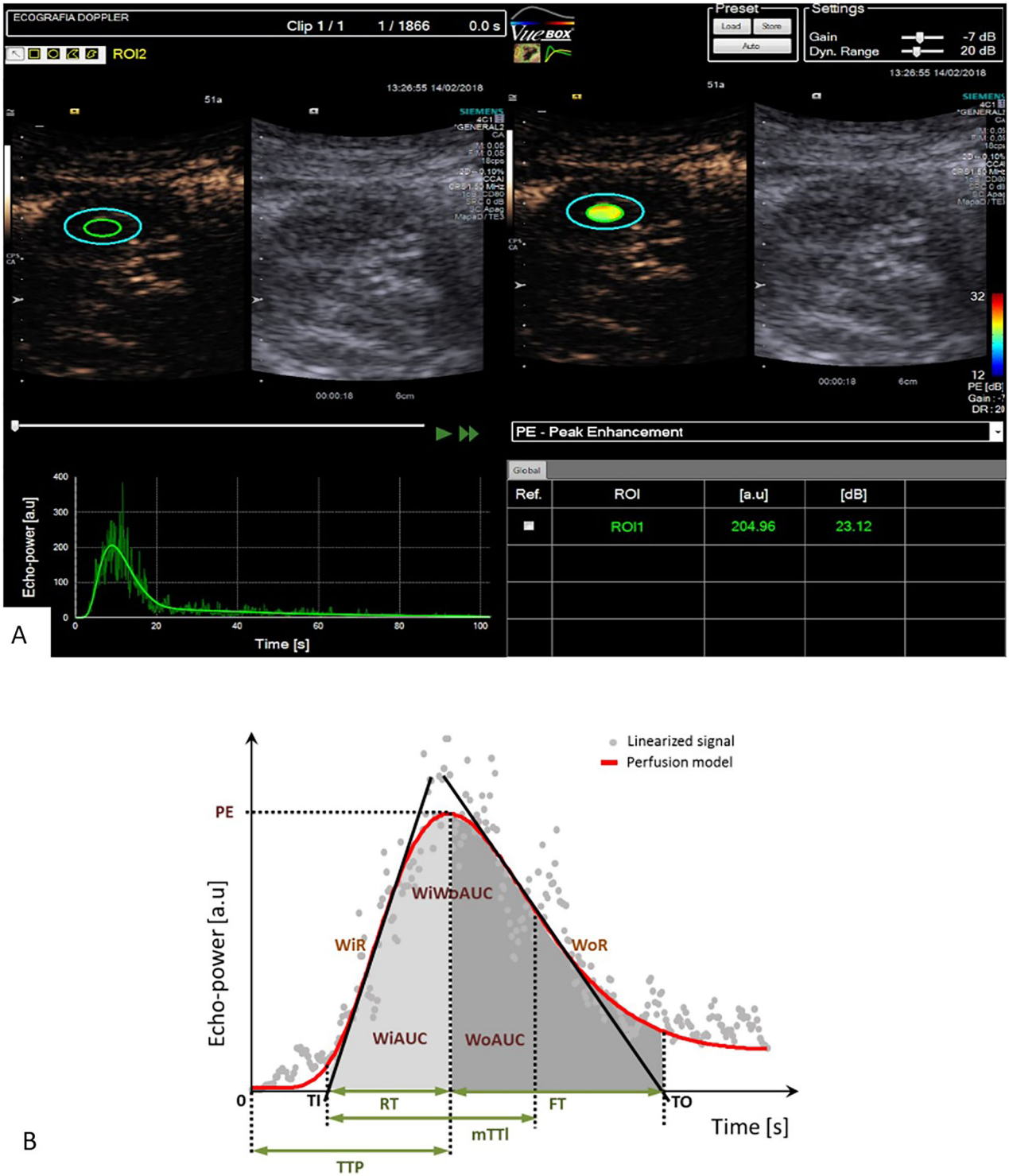


Fig. 1 Dynamic contrast-enhanced ultrasound of a pancreatic graft, obtaining TIC parameters through VueBox software (Bracco®). **A** The region of interest is manually placed in the graft parenchyma, avoiding large vessels, and a TIC and its eleven parameters are shown. **B** Schema of the eleven obtained parameters for each time-intensity-curve: PE (a.u.), WiAUC (a.u.), RT (s), mTTI (s), TTP (s), WiR (a.u.), WiPi (WiAUC/RT, a.u.), WoAUC (a.u.), WiWoAUC (a.u.), fall time (FT, s), WoR (a.u.). Bracco—VueBox® User Manual—Bracco Suisse SA Copyright © 2010 Bracco Suisse SA ALL RIGHTS RESERVED

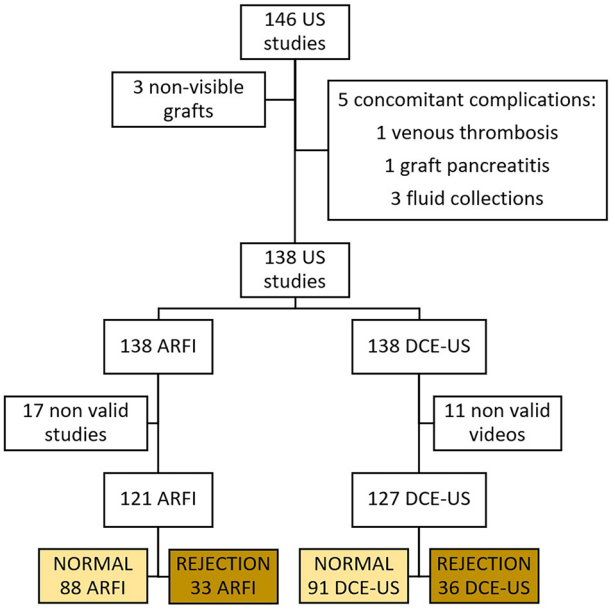


Fig. 2 Patient flow chart showing included and excluded ultrasonographic studies and final group classification according to the presence of rejection. US, ultrasonography; ARFI, acoustic radiation force impulse; DCE-US, dynamic contrast-enhanced ultrasound

establish a diagnosis according to the Banff criteria (2011 revision) [32].

The presence or absence of rejection and fibrosis was recorded in all biopsies. For those cases with rejection, both the type of rejection and its degree were recorded (T cell-mediated, antibody-mediated, and chronic active antibody-mediated rejection).

Statistical analysis

Categorical variables are expressed as absolute frequencies and percentages (%). Quantitative variables are presented as median and interquartile range values (IQR: 25th–75th percentiles).

The results of the US studies were divided into two groups (normal/rejection) according to the biopsy result. Studies without a biopsy (1-week study and non-accessible grafts) or with an inconclusive biopsy result, were classified according to the clinical criteria adopted at the time of the US study. Clinical criteria were defined as > 3-fold increase in amylase and/or lipase, or de novo DNA with pancreas graft dysfunction. Elastography and DCE-US values were compared between groups.

An analysis of the parameters according to the post-transplantation time was performed, dividing them into 3 temporal groups in relation to the three protocolized US studies (0–10 days, 11–90 days, and > 90 days). This was carried out to better evaluate the behavior of the parameters during the postoperative period, since some authors have published variations of the elastography values of liver grafts depending on the time after transplantation [16, 33]. Elastography and DCE-US parameters in normal and rejection groups were compared using Fisher's exact test or the *U* of Mann–Whitney test for categorical and continuous parameters, respectively. A subanalysis including only biopsy-confirmed cases was also performed to avoid the bias introduced by the classification of some US studies without histopathologic confirmation. Odds ratios (OR) and area under the curve (AUC), with their 95% confidence intervals (95% CI), were evaluated using logistic regression models for each radiological parameter in both techniques, using predetermined cut-offs following Youden's criteria, with rejection as the response variable. A combined score for predicting rejection was constructed using elastography, PE, and WiR, as these are the most commonly used parameters in the literature [25, 34–36], particularly in pancreas transplantation [25].

Finally, the US studies with biopsy were divided into two groups according to the presence of graft fibrosis (no fibrosis/fibrosis), and a comparison of elastography and DCE-US parameters was also performed, using the same statistical analysis as explained above.

The significance level was set at 5% (two-sided). Data collection was performed using SPSS v27.0 statistical package (IBM Corp. ®), and all statistical analyses were performed using SAS v9.4 software (SAS® Institute).

Results

A total of 146 US studies in 56 patients were initially included. Three studies could not be performed due to non-optimal visualization of the graft (intestinal loop interposition). A further 5 studies were excluded due to the presence of other complications (1 venous thrombosis, 1 graft pancreatitis, and 3 fluid collections). A total of 138 studies in 56 patients were finally included, as shown in Fig. 2, 98 (71%) in the normal group and 40 (29%) in the rejection group. Patient and graft characteristics of both groups are shown in Table 1.

Of the 88 studies with concomitant biopsy (biopsy-confirmed cases), 48 had no rejection, 13 presented indeterminate changes for rejection (and were classified according to clinical management), and 27 had rejection. Of the rejection-confirmed cases, 26 had acute T-cell-mediated rejection (20 mild/grade I, 5 moderate/grade II, and 1 severe/grade III), 1 had antibody-mediated rejection, and no cases of chronic active antibody-mediated rejection were found.

Elastography (ARFI) to predict rejection

A total of 138 ARFI studies were performed. Seventeen were discarded because they did not meet the quality

Table 1 Patient characteristics of included ultrasonographic studies according to the presence of rejection (normal/rejection)

	Total	Normal	Rejection	*p* value
US included studies (*n*)	138	98	40	N/A
Patient sex (M:F)	72:66	48:50	24:16	0.240
Patient age (median [IQR], years)	40 [33–51]	40 [33–50]	43 [34–52]	0.193
Donor age (median [IQR], years)	39 [28–45] (8 missing)	37 [24–45] (5 missing)	44 [31–45] (3 missing)	0.028
Cold Ischemia time (median [IQR], h)	8.1 [6.4–10.3] (5 missing)	8.1 [6.3–10.2] (3 missing)	8.7 [7–10.9] (2 missing)	0.086
Time after transplantation (median [IQR], days)	46 [17–387]	33 [10–373]	346 [27–804]	0.022
Indication of the study				< 0.001
Surveillance	94 (68.2%)	80 (81.6%)	14 (35%)	
1 week	30	28	2	
3 weeks	34	29	8	
1 year	22	23	4	
Dysfunction	23 (16.7%)	9 (9.2%)	15 (37.5%)	
Rejection follow-up	18 (13.1%)	9 (9.2%)	11 (27.5%)	

US ultrasonography, *N/A* not applicable, *M* male, *F* female, *IQR* interquartile range, *IQR/M* interquartile range/median

Table 2 Results of elastography studies according to the presence of rejection (normal/rejection) and comparison between them

	Total	Normal	Rejection	*p* value
Valid ARFI studies (*n*)	121	88	33	N/A
IQR/M (median [IQR])	0.23 [0.17–0.26]	0.23 [0.18–0.27]	0.21 [0.17–0.26]	0.310
ARFI median (median [IQR], m/s)				
Total; *n*	1.37 [1.00–1.83]; 121	1.26 [0.95–1.83]; 88	1.47 [1.34–1.77]; 33	0.028
0–10 days; *n*	1.82 [1.13–2.23]; 22	1.82 [1.13–2.23]; 22	N/A; 0	N/A
11–90 days; *n*	1.44 [1.10–1.89]; 47	1.35 [1.10–1.89]; 33	1.55 [1.27–1.77]; 14	0.675
> 90 days; *n*	1.24 [0.93–1.49]; 52	0.97 [0.88–1.21]; 33	1.46 [1.34–1.84]; 19	< 0.001
ARFI median in biopsy confirmed studies > 90 days (median [IQR], m/s)	1.34 [0.95–1.55] ;37	0.97 [0.89–1.37]; 21	1.47 [1.39–1.91]; 16	< 0.001

ARFI acoustic radiation force impulse, *N/A* not applicable, *IQR* interquartile range, *IQR/M* interquartile range/median

criterion, leaving a total of 121 valid ARFI studies: 88 in the normal group and 33 in the rejection group, as shown in Table 2.

Significant differences were found when comparing the median ARFI of both groups, with a higher stiffness in the rejection group (1.26 m/s vs 1.47 m/s, $p = 0.028$). When analyzing the variation of elastographic values over time (Fig. 3A), a very high dispersion of the values was observed in the immediate and early postoperative periods (0–10 and 11–90 days), which was attributed to the initial local inflammatory changes associated with surgery. When grafts older than 90 days were evaluated (Fig. 3B), a lower dispersion of the values was noted, with more evident differences between groups (0.97 m/s vs 1.46 m/s, $p < 0.001$). Using Youden's criteria, the cut-off value of 1.27 m/s was considered optimal for predicting rejection in grafts > 90 days, with an OR of 16.92 (4.59-62.37, 95% CI) and an AUC of 0.80 (0.70–0.90, 95% CI).

In the subanalysis of the biopsy-confirmed studies in grafts older than 90 days (21 in the normal group and 16 in the rejection group, Table 2), the previously obtained results were confirmed (0.97 m/s vs 1.47 m/s, $p < 0.001$).

Dynamic contrast-enhanced ultrasound (DCE-US) to predict rejection

Of the 138 initial studies, 11 were discarded because of uncorrectable movement in the recorded video [31], leaving a total of 127 valid DCE-US: 91 in the normal group and 36 in the rejection group (Fig. 2).

When the 11 DCE-US parameters were initially compared between groups, no significant differences were found (data presented in the Supplementary Material). Similar to the elastography studies, a very high dispersion of all DCE-US values was observed in the immediate and early postoperative periods (graphics included in the Supplementary Material), making them less reliable. After excluding the values of the first 90 days post-transplantation (Table 3), significant differences were found in the 4 AUC-related parameters showing lower perfusion in the rejection group (WiAUC: 3387 vs 1589

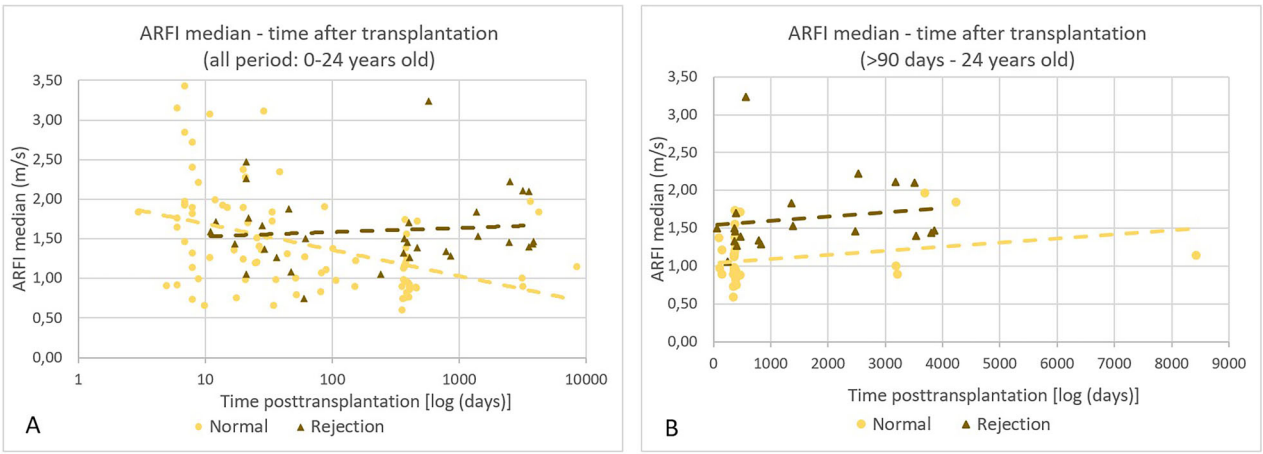


Fig. 3 Elastography median values and their variation depending on time after transplantation. **A** Including all performed studies. **B** Including only grafts > 90-days-old. ARFI, acoustic radiation force impulse

Table 3 Results of DCE-US studies according to the presence of rejection (normal/rejection) and comparison between them

	Total	Normal	Rejection	*p* value
Valid DCE-US studies (*N*)	127	91	36	N/A
QoF, median [IQR]; %	88 [83–93]	88 [81–93]	88 [84–93]	0.595
DCE-US parameters (> 90 days)				
N	50	31	19	N/A
PE, median [IQR]; a.u.	593 [362–954]	656 [425–1101]	439 [258–776]	0.058
WiAUC, median [IQR]; a.u.	2750 [1323–4599]	3387 [1854–7421]	1589 [1045–2748]	0.008
RT, median [IQR]; s	6.3 [5.3–8.6]	7 [5.3–9.3]	5.9 [5.1–7.5]	0.276
mTTI, median [IQR]; s	28.3 [22.1–41]	29.1 [22.7–41.8]	26.2 [17.9–30.8]	0.181
TTP, median [IQR]; s	10.8 [8.5–13.3]	11.5 [8–13.8]	10.3 [7.7–12.1]	0.223
WiR, median [IQR]; a.u.	118 [94–185]	126 [96–185]	110 [56–223]	0.134
WiPI, median [IQR]; a.u.	372 [230–630]	443 [269–694]	274 [164–491]	0.044
WoAUC, median [IQR]; a.u.	5106 [2434–7772]	5789 [2634-13,887]	2828 [1880–5181]	0.019
WiWoAUC, median [IQR]; a.u.	7783 [3899–12,316]	9138 [4259–19,722]	4628 [2925–8933]	0.020
FT, median [IQR]; s	11.3 [9.6–16.5]	13.3 [9.9–24.1]	10.9 [9–11.9]	0.238
WoR, median [IQR]; a.u.	50 [22–69]	48 [22–69]	50 [20–94]	0.617
DCE-US parameters in biopsy-confirmed studies > 90 days:				
N	37	20	17	N/A
PE, median [IQR]; a.u.	508 [329–829]	708 [409–990]	406 [258–509]	0.018
WiAUC, median [IQR]; a.u.	2036 [1229–3613]	3459 [1753–6372]	1569 [1045–2037]	0.002
RT, median [IQR]; s	5.9 [5–8.1]	6.7 [5–8.8]	6 [5.4–6.5]	0.686
mTTI, median [IQR]; s	28.4 [20.3–41]	32.2 [22–41.7]	26.4 [18.2–30.8]	0.408
TTP, median [IQR]; s	10.4 [8.2–13]	10.7 [8.2–13.2]	10.4 [8.4–12.1]	0.648
WiR, median [IQR]; a.u.	115 [91–160]	124 [95–199]	105 [55–116]	0.065
WiPI, median [IQR]; a.u.	316 [208–515]	467 [259–676]	266 [163–316]	0.015
WoAUC, median [IQR]; a.u.	3530 [2088–7363]	6197 [2480–12,524]	2818 [1880–3766]	0.008
WiWoAUC, median [IQR]; a.u.	5564 [3287–10,295]	9607 [3772–19,586]	4408 [2924–5803]	0.006
FT, median [IQR]; s	10.9 [9.4–15.6]	10.9 [9.5–17.5]	10.9 [9–11.7]	0.860
WoR, median [IQR]; a.u.	88 [82–92]	46 [23–93]	49 [20–56]	0.146

DCE-US dynamic contrast-enhanced ultrasonography, *N* number of included studies, *N/A* not applicable, *QoF* quality of fit, *IQR* interquartile range, *IQR/M* interquartile range/median, *PE* peak enhancement, *WiAUC* wash-in area under the curve, *RT* rise time, *mTTI* mean transit time, *TTP* time to peak, *WiR* wash-in rate, *WiPI* wash-in perfusion index (WiAUC/RT), *WoAUC* wash-out area under the curve, *WiWoAUC* total area under the curve, *FT* fall time, *WoR* wash-out rate

Table 4 Cut-off values of the elastography and DCE-US parameters for predicting rejection in grafts older than 90 days

	Cut-off value	OR (95% CI)	*p* value	AUC (95% CI)	Sensibility (%)	Specificity (%)	PPV (%)	NPV (%)
ARFI median (m/s)	1.27	16.92 (4.59–62.37)	< 0.001	0.80 (0.70–0.90)	89.47	78.79	70.83	92.86
PE (a.u.)	601	0.25 (0.09–0.72)	0.010	0.67 (0.55–0.79)	73.68	64.55	56.00	80.00
WiAUC (a.u.)	2748	0.19 (0.06–0.55)	0.002	0.70 (0.58–0.82)	78.95	67.74	60.00	84.00
RT (s)	4.9	1.64 (0.44–6.16)	0.461	0.54 (0.44–0.63)	21.05	80.65	40.00	62.50
mTTI (s)	15.6	1.56 (0.26–9.23)	0.622	0.52 (0.45–0.59)	10.52	93.54	50.00	63.04
TTP (s)	11.4	0.38 (0.13–1.08)	0.069	0.62 (0.50–0.74)	68.42	51.61	46.43	72.73
WiR (a.u.)	118	0.28 (0.10–0.81)	0.018	0.65 (0.53–0.77)	73.68	61.29	53.85	79.17
WiPI (a.u.)	369	0.25 (0.09–0.72)	0.010	0.67 (0.55–0.79)	73.68	64.52	56.00	80.00
WoAUC (a.u.)	5181	0.21 (0.07–0.62)	0.005	0.69 (0.57–0.80)	78.95	64.52	57.69	83.34
WiWoAUC (a.u.)	6388	0.22 (0.08–0.64)	0.006	0.68 (0.56–0.80)	73.68	67.74	58.34	80.77
FT (s)	12.9	0.42 (0.15–1.21)	0.108	0.60 (0.48–0.72)	78.95	54.84	51.72	80.95
WoR (a.u.)	57	0.59 (0.21–1.63)	0.307	0.56 (0.44–0.69)	68.42	41.94	41.94	68.42

ARFI acoustic radiation force impulse, *PE* peak enhancement, *WiAUC* wash-in area under the curve, *RT* rise time, *mTTI* mean transit time, *TTP* time to peak, *WiR* wash-in rate, *WiPI* wash-in perfusion index (WiAUC/RT), *WoAUC* wash-out area under the curve, *WiWoAUC* total area under the curve, *FT* fall time, *WoR* wash-out rate, *AUC* area under the curve, *PPV* positive predictive value, *NPV* negative predictive value

a.u., $p = 0.008$; WiPi: 443 vs 274 a.u., $p = 0.044$; WoAUC: 5789 vs 2828 a.u., $p = 0.019$; WiWoAUC: 9138 vs 4628 a.u., $p = 0.020$). Time-dependent parameters (RT, mTTI, TTP, and FT) and WoR were not useful in predicting rejection according to these criteria.

Using Youden's criteria in grafts older than 90 days, optimal cut-off values were found for the previously mentioned AUC parameters and also for PE and WiR, as shown in Table 4. It is worth noting that all of them have a negative predictive value > 79%. DCE-US parameters obtained lower AUC values (0.67–0.70) than ARFI (0.80).

In the subanalysis of the biopsy-confirmed studies in grafts older than 90 days (20 in the normal group and 17 in the rejection group, Table 3), the previously obtained results were confirmed (WiAUC: 3459 vs 1569 a.u., $p = 0.002$; WiPi: 467 vs 266 a.u., $p = 0.015$; WoAUC: 619 vs 2818 a.u., $p = 0.008$; WiWoAUC: 9607 vs 4408 a.u., $p = 0.006$), also with statistical differences in PE (708 vs 406 a.u., $p = 0.018$).

ARFI and DCE-US predictive score for rejection

The combined predictive score for rejection was constructed according to the following rules: 0 points when ARFI ≤ 1.27, 1 point if ARFI > 1.27 and PE ≥ 601 and WiR ≥ 118, and 2 points for the rest (ARFI > 1.27 and PE < 601 or ARFI > 1.27 and WiR < 118). The higher the score (0, 1, or 2), the greater the likelihood of rejection. To assess the adjustment of this score, OR and AUC with 95% CI were also estimated using a trend test (1 unit increase), and specific contrast parameters between categories were performed, as shown in Table 5. A calibration plot was created to visually assess the goodness of fit between observed and predicted probabilities of the event (Fig. 4). Following this score, a graft with alteration of both elastography and DCE-US values (2 points) has a 23.2-fold probability of rejection compared to a graft with normal ARFI and DCE-US studies (0 points).

Table 5 Combined predictive score for diagnosing rejection and the OR associated with each category contrast

	Score
ARFI ≤ 1.27	0
ARFI > 1.27 and PE ≥ 601 and WiR ≥ 118	1
ARFI > 1.27 and (PE < 601 or WiR < 118)	2

Score	OR (95% CI)	*p* value	AUC (95% CI)
0 vs 1	9.06 (1.69–48.61)	< 0.0001	0.82 (0.72–0.92)
0 vs 2	23.2 (5.44–98.86)		

ARFI acoustic radiation force impulse, *PE* peak enhancement, *WiR* wash-in rate, *OR* odds ratio, *AUC* area under the curve, *CI* confidence interval

ARFI and DCE-US to predict graft fibrosis

Of the 88 biopsy samples (all periods), 62 had no fibrosis (70.5%) and 26 presented fibrosis (22 mild, 3 moderate and 1 severe). The group with no fibrosis presented younger grafts than the group with fibrosis (time post-transplantation: 61 vs 384 days, $p = 0.041$).

Comparison of elastography and DCE-US values between groups showed no significant differences in any parameter when evaluating the entire period studied (not shown, see electronic supplementary material). Again, an initial high dispersion of both elastography and DCE-US values was noted in the first 90 days post-transplantation. When including only grafts > 90 days after transplantation, the fibrosis group showed higher stiffness (1.15 vs

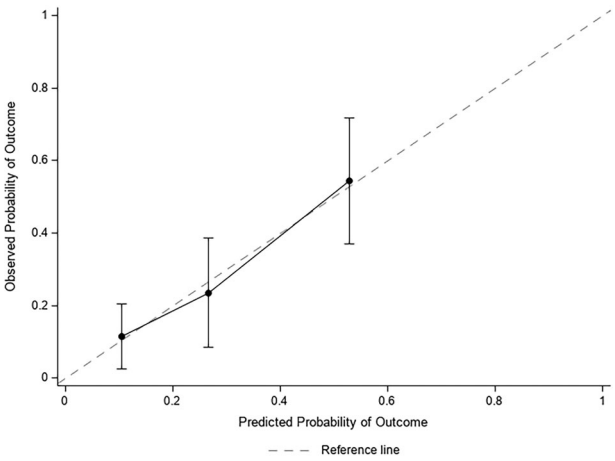


Fig. 4 Calibration plot of the predictive combined score, to assess the goodness between observed and predicted probabilities of the event

1.42 m/s, $p = 0.027$) and lower graft perfusion (WiAUC 3156 vs 1569 a.u., $p = 0.023$; WoAUC 5444 vs 2828 a.u., $p = 0.041$, and WiWoAUC 8697 vs 4628 a.u., $p = 0.040$). The remaining DCE-US parameters showed no significant differences between the groups.

Discussion

We report the first prospective study to evaluate the role of elastography, DCE-US and the combination of both in the diagnosis of rejection in pancreas transplantation. It reveals a higher stiffness and lower perfusion in grafts with rejection after the initial postsurgical period (first 90 days). We have also assessed for the first time the normal elastography and DCE-US values of the pancreas graft.

Both techniques presented a high dispersion of the values during the initial postoperative period (0–90 days), which could be attributed to local inflammatory and vascular postsurgical factors. This is comparable to what has been previously reported in the elastography of liver grafts [16, 33]. For this reason, elastography and DCE-US studies during this initial period may be unreliable and should be carefully interpreted.

After the first 90 days post-transplantation, when dispersion of the values decreases, the elastography values of the normal grafts are similar to those published on healthy native pancreas (1.0 ± 0.2 m/s) [37]. In the presence of rejection, a significantly higher stiffness of the graft is found. The main histological features of acute t-cell mediated rejection (which is the main type of rejection in our series) correspond mainly to inflammation of the different components of the pancreatic graft, which is generally caused by a genetic disparity between the donor and recipient, mainly affecting the fibrous septa, vessels, ducts and acini [32]. Therefore, we can infer that graft inflammation results in higher pancreas graft stiffness, which has also been reported in liver [16, 33] and kidney [17] graft rejection. However, inconsistent results have been reported in kidney grafts, Kim et al [24], found no differences in the elastography of renal grafts with or without rejection, although they only included normofunctioning grafts (to predict subclinical rejection), reducing the possibility of finding statistically significant differences. In relation to DCE-US, less perfusion is seen in grafts with rejection after the initial 90 days post-transplantation. In a study published in 2013, Kersting et al [25] found similar results in 14 pancreas transplant patients (23 normal studies and 9 with rejection), reporting lower PE and WiR in the rejection group. The impact of rejection in DCE-US has also been studied in kidney transplantation, Kim et al [24] reported lower perfusion of grafts with rejection, with statistical differences in PE, WiAUC, WiPi, WoAUC, and WiWoAUC, in a similar way to our results. The combination of elastography and DCE-US offers promising results in predicting rejection as reflected in the combined score, improving their individual performance.

The high negative predictive values of the cut-off elastography and DCE-US values found in our study (all above 79%) indicate the strength of these tests when negative, making them most useful when screening normofunctioning grafts and in obtaining a negative result, avoiding an unnecessary surveillance biopsy.

It should be noted that none of these techniques specifically assesses acute rejection, as other pathological conditions of the native pancreas (such as pancreatitis) have also been associated with higher stiffness of the pancreatic tissue [15, 38]. Moreover, as demonstrated in our study, fibrosis also implies a higher stiffness and lower perfusion of the grafts > 90 days post-transplantation. Graft fibrosis correlates with graft survival, its progression over time is well documented, particularly in patients with repeated acute rejection episodes and especially in chronic rejection [32, 39], not represented in this study. It is present in other causes of graft dysfunction as documented in a recent meta-analysis of elastography in kidney transplantation [40]. Yoon et al [16] found that elastography in liver grafts could not discriminate between acute rejection, chronic rejection, and fibrosis, which may overlap, and proposed a combination of elastography and biochemical tests to correctly distinguish these different entities. Similarly, any condition causing pancreas graft edema or congestion, such as vascular complications, pancreatitis, and infection, would impair graft perfusion, making it necessary to interpret the DCE-US and elastography values in the clinical context of the patient and, ideally, confirm a suspected rejection with a percutaneous US-guided biopsy.

Both techniques, elastography and DCE-US, are non-invasive and available in most centers with a transplantation program, making them easy to integrate into daily practice. Furthermore, both techniques do not require the entire pancreas graft to be visible, as they can be carried out in a representative part of the parenchyma (in our series, only three grafts were not visible at all). To obtain reliable results, they should be performed by personnel experienced in advanced US techniques. US elastography is the easiest technique to perform as it is faster and does not require peripheral access for injection. Nevertheless, the US technique has certain limitations. First, compression with the probe can artificially increase tissue stiffness, leading to an overestimation of elastography values. Therefore, it is essential that the operator avoids exerting pressure on the area being examined. Second, in some cases, the graft cannot be adequately visualized due to its deep anatomical location, resulting in suboptimal studies that fail to meet quality standards; in our series, 17 examinations were excluded for this reason. These issues could potentially be addressed using MRE, a less widely available and more costly technique that has not yet been studied in the context of pancreas transplantation. DCE-US requires a peripheral line, contrast administration, and several acquisition and post-processing steps (recording a 2-min video without significant movement, good ROI selection, good signal-to-noise ratio, and good QoF of the TIC). By looking at our results, we suggest that using only wash-in parameters may simplify this process, as the wash-in phase usually takes less than 15 s (reducing the probability of technical errors) and contains most of the useful parameters studied. This study has some limitations. First, not all the US studies performed have pathological confirmation. Nevertheless, the post hoc subanalysis, including only the biopsy-confirmed studies, validated the previous results. Second, although the acquisition conditions have been standardized, certain physiological parameters, such as blood pressure or cardiac output, remain beyond the operator's control and may act as confounding factors in DCE-US studies. This could explain discrepancies and unexpected findings observed across different sessions during follow-up, despite using an identical standardized approach [30]. Third, the unicentric nature of the study and the limited number of patients are also limitations, and further prospective multicentric studies with a large number of patients are needed to confirm our findings.

Despite these inconveniences, the ability of elastography and DCE-US to detect changes in pancreas grafts, the non-invasive nature of these tools (with no radiation or nephrotoxic contrast agents) and their easy integration into the daily US routine, make them promising novel tools to aid in the diagnosis of acute cellular pancreatic graft rejection.

Abbreviations

ARFI	Acoustic radiation force impulse
AUC	Area under the curve
CEUS	Contrast-enhanced ultrasound
DCE-US	Dynamic contrast-enhanced ultrasound
MRE	Magnetic resonance elastography
mTTI	Mean transit time
OR	Odds ratio
PE	Peak enhancement
QoF	Quality of fit
RT	Rise time
TIC	Time-intensity curve
TTP	Time to peak
US	Ultrasonography
WiAUC	Wash-in area under the curve
WiPI	Wash-in perfusion index
WiR	Wash-in rate
WiWoAUC	Wash-in and wash-out area under the curve
WoAUC	Wash-out area under the curve
WoR	Wash-out rate

Supplementary information

The online version contains supplementary material available at https://doi.org/10.1007/s00330-025-11772-1.

Funding

This study has received funding from a grant from the Spanish Ministry of Health (FIS 16/00167).

Compliance with ethical standards

Guarantor

The scientific guarantor of this publication is Ángeles García-Criado (Hospital Clínic de Barcelona).

Conflict of interest

The authors of this manuscript declare no relationships with any companies, whose products or services may be related to the subject matter of the article.

Statistics and biometry

One of the authors (V.S.) has significant statistical expertise and has provided statistical advice for this manuscript.

Informed consent

Written informed consent was obtained from all subjects (patients) in this study.

Ethical approval

Institutional Review Board approval was obtained.

Study subjects or cohorts overlap

N/A.

Methodology

- Prospective
- Experimental
- Performed at one institution

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Received: 22 January 2025 Revised: 26 April 2025 Accepted: 22 May 2025
Published online: 25 June 2025

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